



Fabio Vulpi

Ph.D. in Mechanical Engineering and Management

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ABOUT ME

Daring and versatile, I firmly believe that attaining successful outcomes relies on both the ability to ask pertinent questions and the commitment to finding the answers.

SOFT-SKILLS

- Pragmatic
- Team working
- Result Oriented

DIGITAL SKILLS

Physic Simulation

MATLAB | Ansys FLUENT
Simulink | Wolfram Mathematica

µC programming

Debian | Python | C | Arduino
Linux | ROS | C++ | C#

3-D Modeling

AutoCAD 2D | Autodesk Inventor
SolidWorks | Cura (Ultimaker)
MeshLab | Eagle - PCB design

LANGUAGES

Italiano
English
Deutsch

C 2
C 1
A 2

PUBLICATIONS

- Kalman Supervised Network for improved Model Predictions.
- Recurrent and convolutional NN for deep terrain classification by autonomous robots.
- An RGB-D multi-view perspective for autonomous agricultural robots.
- On the role of feature and signal selection for terrain learning in planetary exploration.
- Usability Study of WISE for Movement Assessment and Exercise.
- Wearable Inertial Sensors for Range of Motion Assessment (WISE).

WORK EXPERIENCE

Experienced Consultant 10/2024 - current

Amaris Consulting - Turin (IT)

Vehicle Functional Safety (FuSa) consultant for ISO 26262:2018. Partnered with the FuSa Manager on the Safety Concept and Development Interface Agreement (DIA), led Hazard Analysis and Risk Assessment (HARA) and Failure Modes and Effects Analysis (FMEA) reviews. Validated ISO-compliant safety requirements throughout the automotive safety lifecycle.

Software System Engineer 01/2023 - 10/2024

Bosch CVIT - Bari (IT)

Served as a Functional Safety Software Engineer on the top-level V-model for Engine and Battery Control Unit development—ensuring ISO 26262 compliance across light-duty and heavy-duty platforms, diesel and electric powertrains. Engineered AI and ML driven vehicle convenience features (such as passive keyless entry and gesture-based trunk release) leveraging advanced signal processing and device-tracking models.

Software Automation Engineer 11/2022 - 01/2023

Links Management & Technology - Bari (IT)

Engineered Artificial Intelligence and Machine Learning software solutions to optimize automation workflows.

EDUCATION

Ph.D. in Mechanical Engineering and Management 11/2019 - 04/2023

Polytechnic of Bari - Bari (IT)

Collaborated with CNR-STIIMA on EU-funded projects ADE, ATLAS, and ANTONIO, leading HW/SW development and AI/ML algorithm design for mechanical systems.

- ADE: Implemented deep-learning terrain-classification pipelines for autonomous exploration rovers.
- ATLAS: Engineered a ROS-based, multi-sensor robotic platform for 3D reconstruction and analysis of agricultural fields.
- ANTONIO: Developed advanced signal- and image-processing workflows for crop-monitoring automated guided robots (AGRs).

Master of Science in Mechanical Engineering 10/2017 - 10/2019

NYU - New York (USA)

110L/110 - GPA 3.952/4

Completed a double-degree program in Mechanical Engineering, specializing in Mechatronics and Robotics, in collaboration with the Polytechnic of Bari. Authored an M.S. thesis in Medical Engineering field on wearable inertial-sensor systems for quantitative assessment of upper-extremity range of motion.

Bachelor Degree in Mechanical Engineering 07/2014 - 07/2017

Polytechnic of Bari - Bari (IT)

109/110

Performed physics-based modeling and numerical computation to investigate the nonlinear dynamics of oscillators with a bistable potential.

AWARDS

Ingeniarius Ltd - Porto (PT) RobotCraft First Place Award - 08/2022

The competition challenged me to design and assemble an autonomous mobile robot that would fully explore a maze and generate LiDAR-based SLAM maps in ROS, then re-navigate the maze optimizing the path using its computed map. I developed real-time C++ and Python algorithms for mapping and path planning and integrated Arduino-driven motor control.